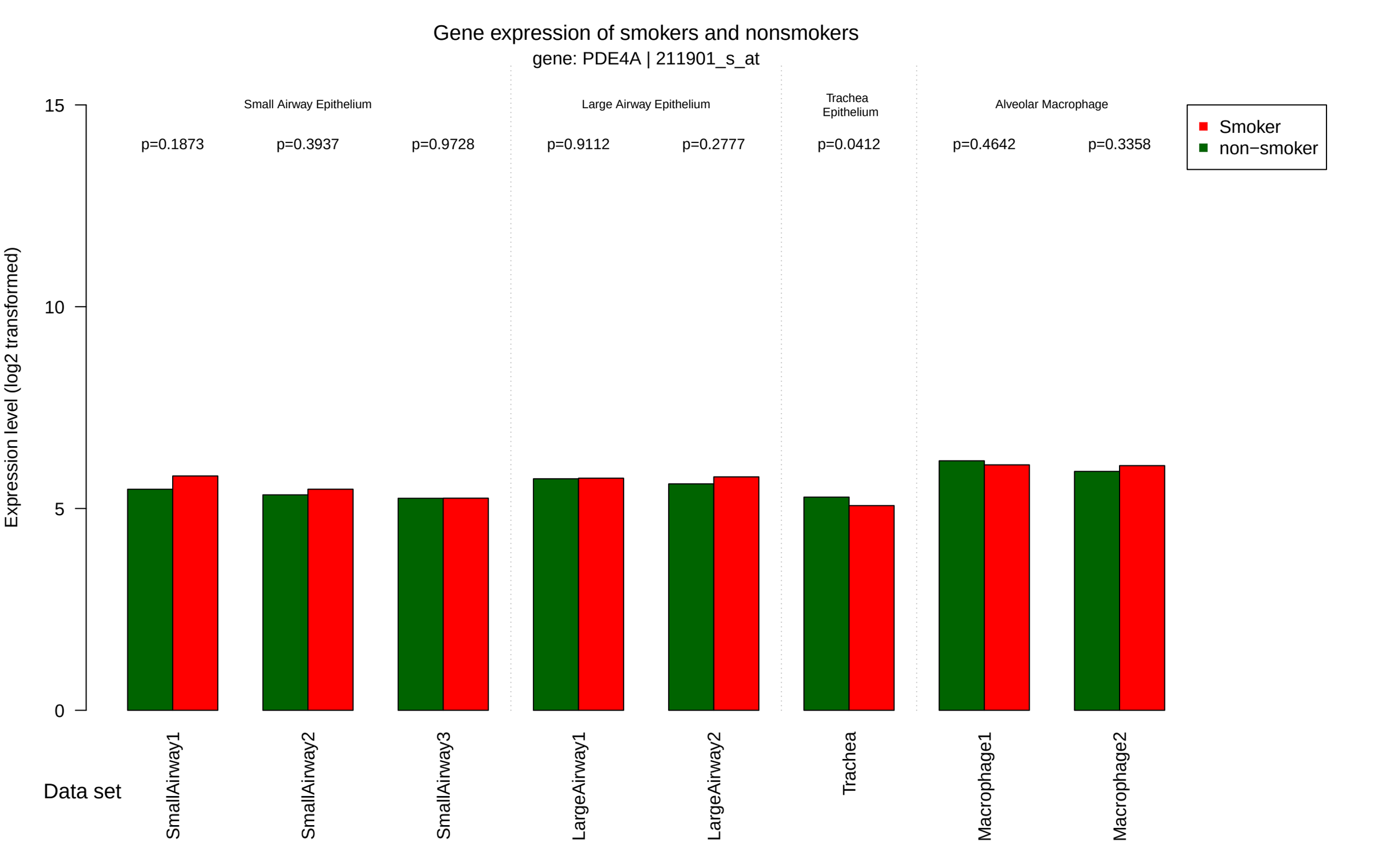
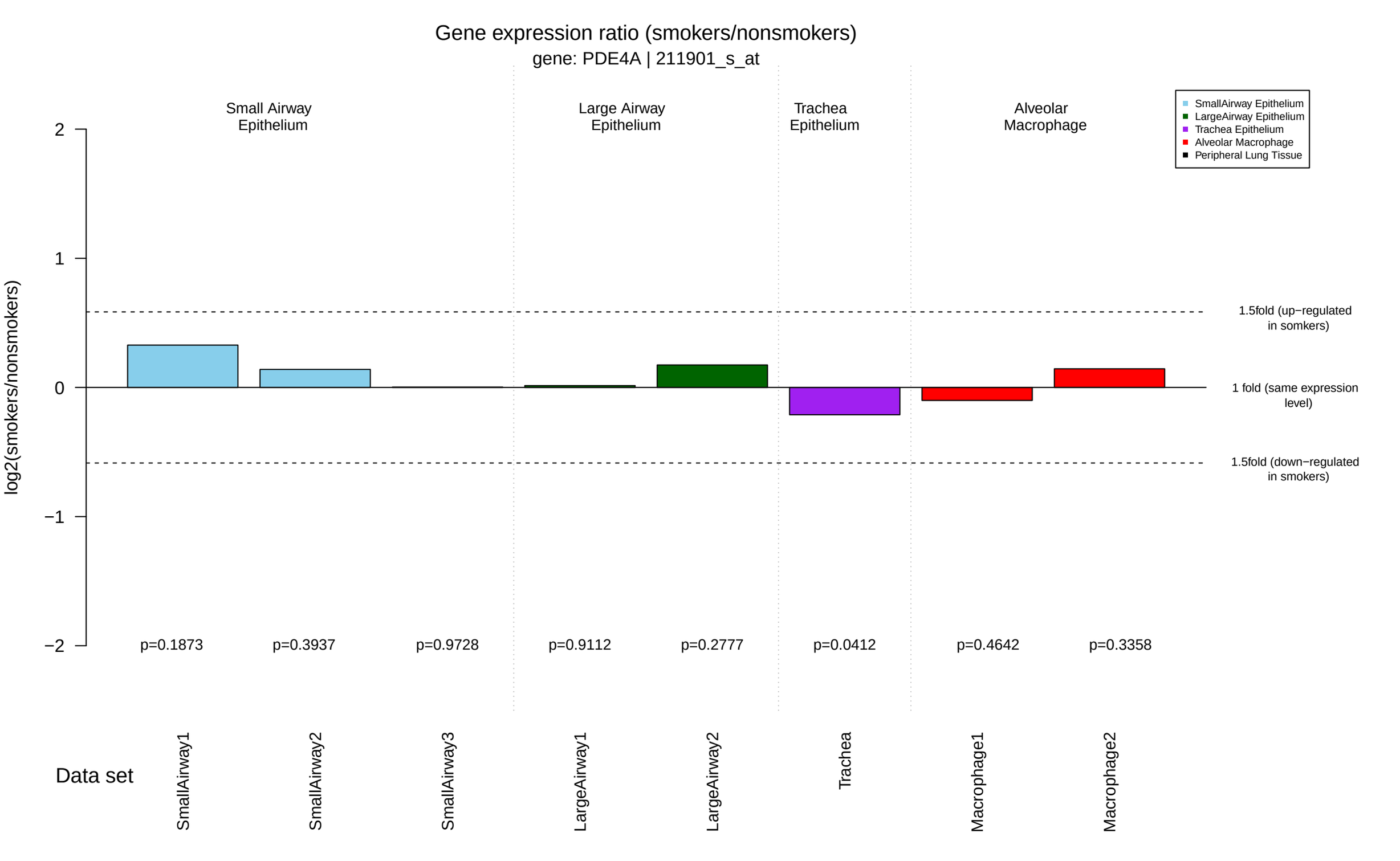
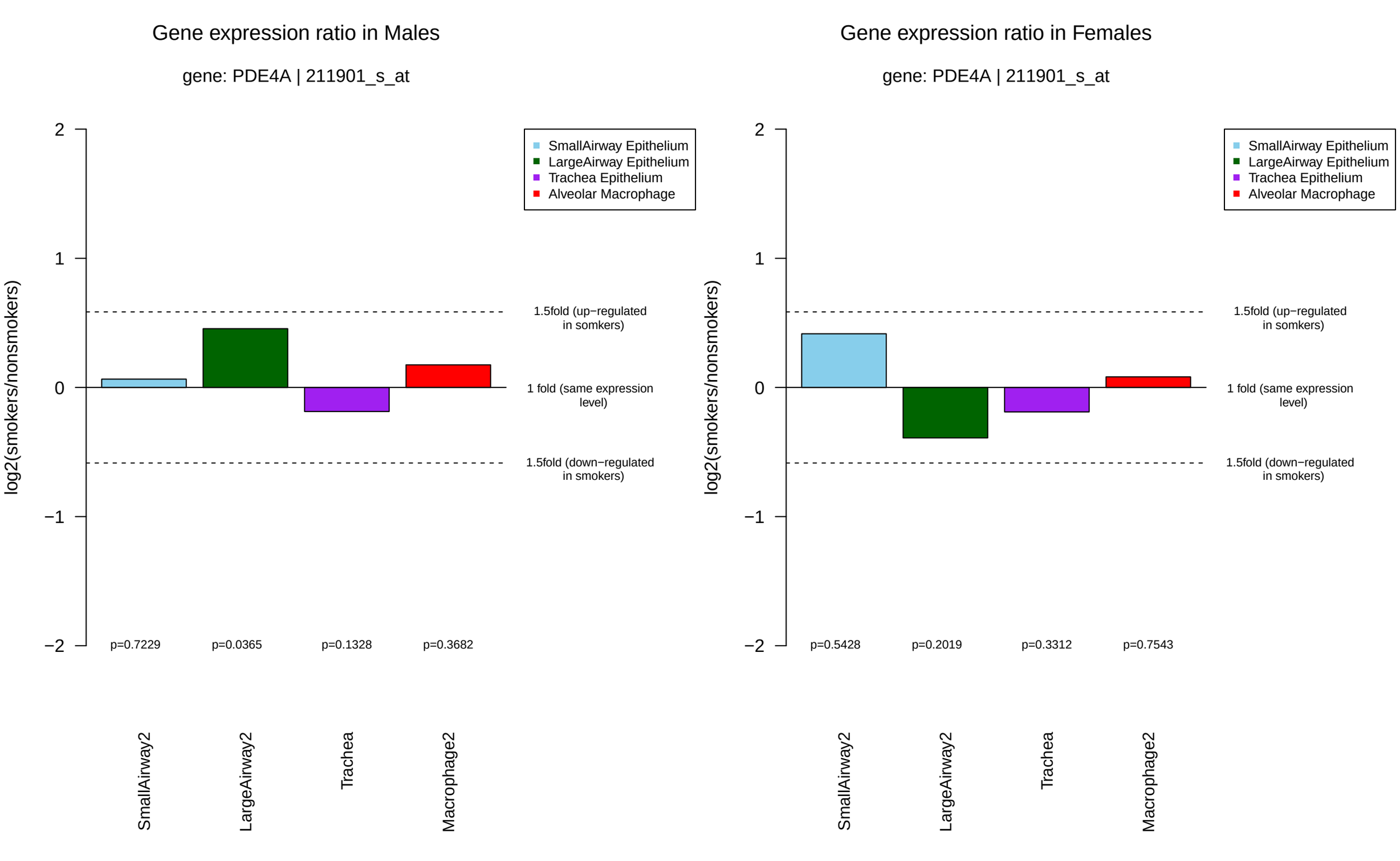
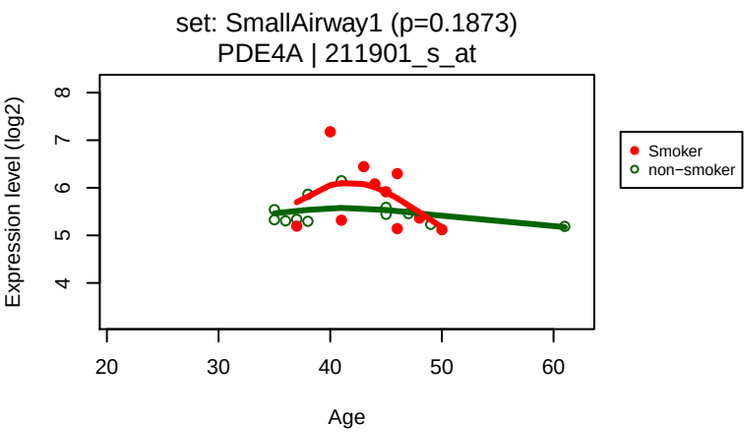
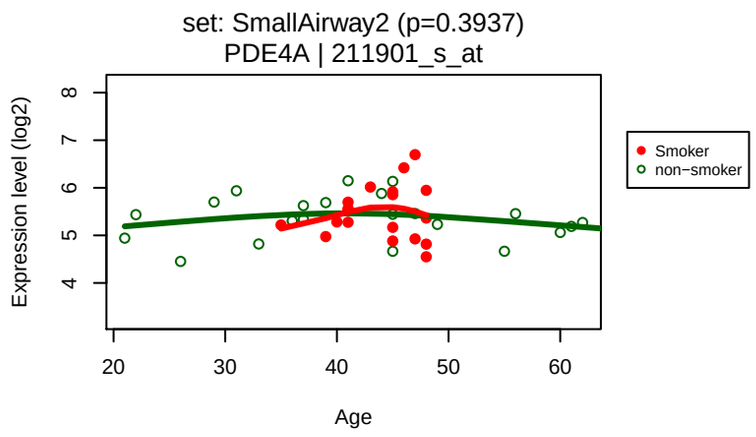
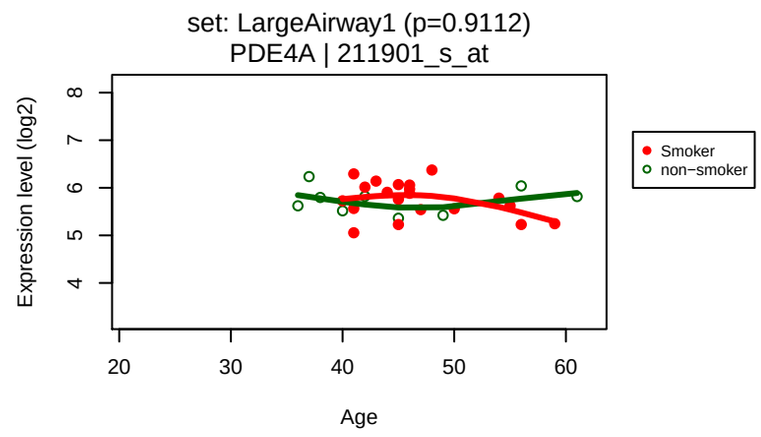

Gene expression of smokers and nonsmokers
gene: PDE4A | 211901_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1873
p=0.3937
p=0.9728
p=0.9112
p=0.2777
p=0.0412
p=0.4642
p=0.3358
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

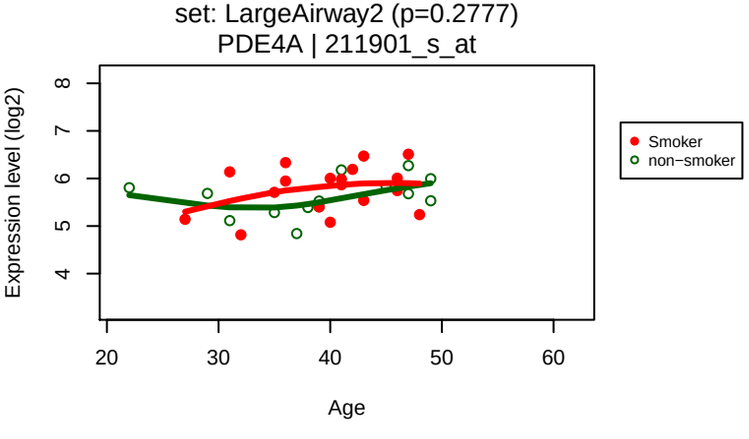

Gene expression ratio (smokers/nonsmokers)
gene: PDE4A | 211901_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1873
p=0.3937
p=0.9728
p=0.9112
p=0.2777
p=0.0412
p=0.4642
p=0.3358
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

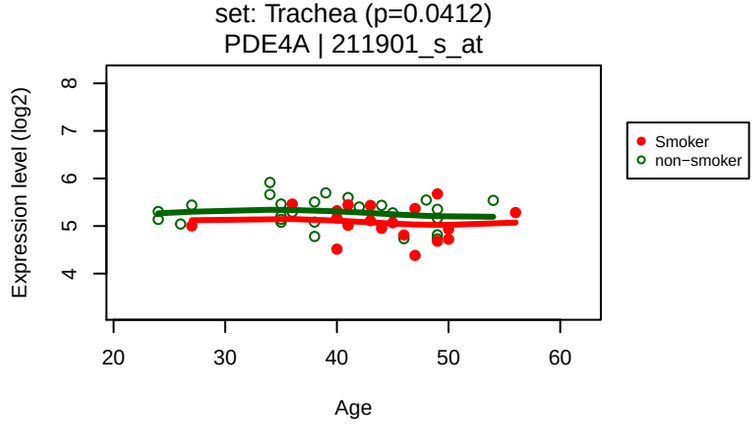

Gene expression ratio in Males
gene: PDE4A | 211901_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7229
p=0.0365
p=0.1328
p=0.3682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE4A | 211901_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5428
p=0.2019
p=0.3312
p=0.7543
SmallAirway2
LargeAirway2
Trachea
Macrophage2

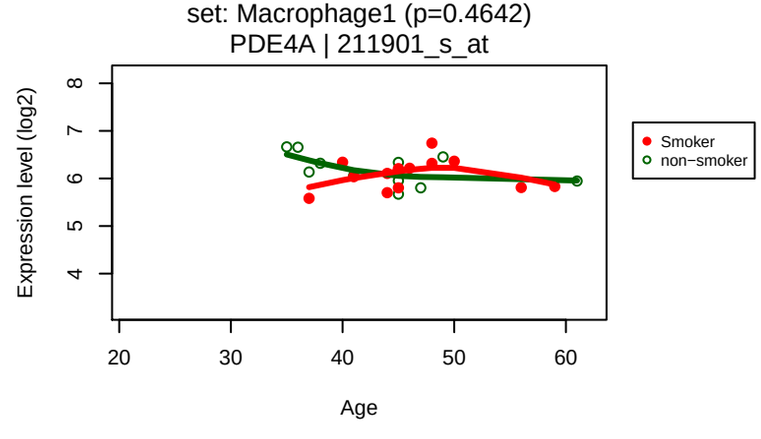

set: SmallAirway1 (p=0.1873)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker

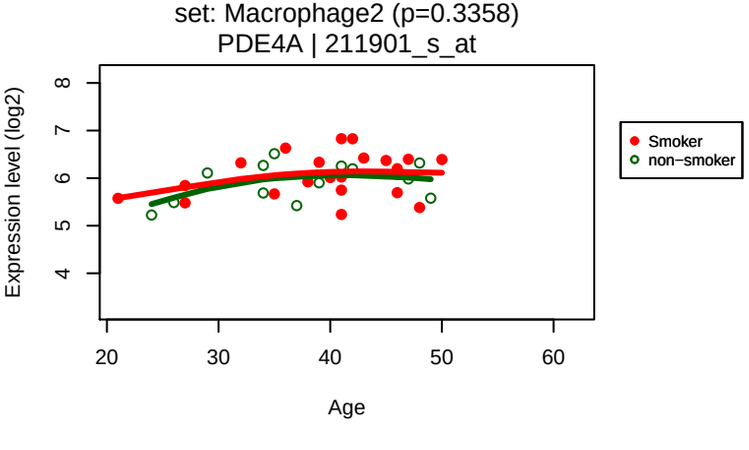

set: SmallAirway2 (p=0.3937)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker

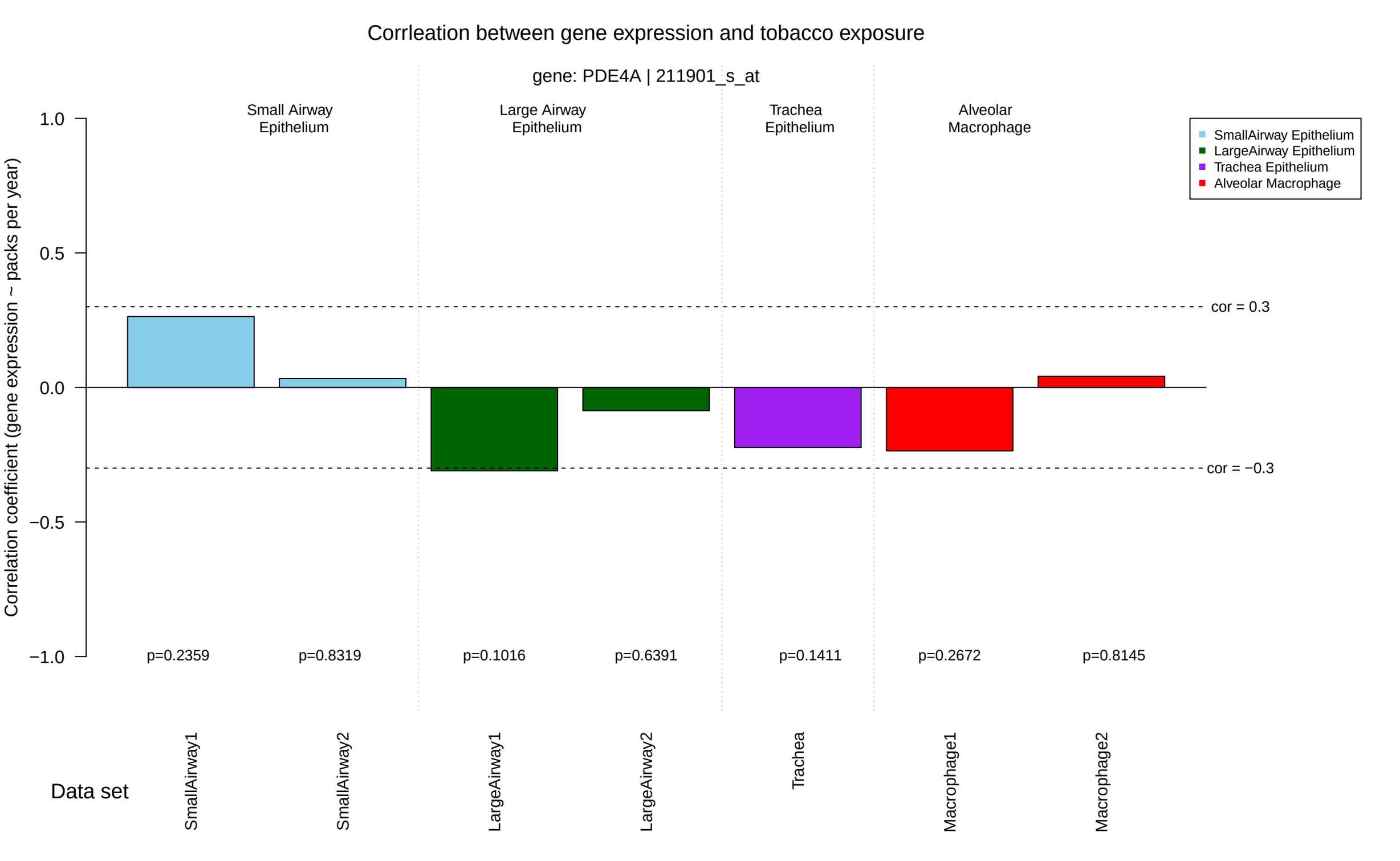

set: LargeAirway1 (p=0.9112)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2777)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0412)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4642)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3358)
PDE4A | 211901_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE4A | 211901_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2359
p=0.8319
p=0.1016
p=0.6391
p=0.1411
p=0.2672
p=0.8145
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2